



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: CS3251 & Programming in C

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

• **Unit / Topic:** Unit V / Sequential & Random Access files

• **Course Outcome:** CO5

• **Topic Learning Outcome:** TLO12

• **Activity Chosen:** Flipped Classroom

• **Justification:**

The flipped classroom teaching method inverts traditional classroom instruction by having students engage with new material (like lectures or readings) outside of class time, and then using class time for activities like discussions, problem-solving, and application of learned concepts. This shift allows for more active and personalized learning experiences, moving away from passive listening in class. Flipped classrooms allow for more interaction between students and instructors, fostering a more collaborative learning environment.

• **Time Allotted for the Activity:** 20 Minutes

• **Details of the Implementation:**

- The time allotted for each topic is 10 Minutes.
- Totally 2 students delivered the content by learning outside their class hours.
- Sequential and Random Access files topics were given to the students.
- As they already learned about this topic in Python many students shown interest to prepare the same using C concepts. Among them only 2 students were selected to present the contents on the mentioned time.
- Students prepared power-point presentation for the given topics and delivered it to their classmates.
- The students presented the topic and the sample of activity is shown in Figure 1 and 2, each for one topic.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	3	2	1	1	1	1	1	1	1	3	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	1	1	1	1	1	3	1	1
Justification for Correlation	Students will be able to apply basic knowledge of engineering in different file operations.	Students will be able to identify relevant file handling methods such as sequential or random file access for the given applications	Students will be able to interpret the complex engineering problems for data management with the help of files and provide solution efficiently.	Students will be able to apply the knowledge on interpretation and retrieval of data using file handling concepts.	Students will be able to use DEV C++ software / Linux to provide solution of the various real time problems	Students will complete the assignment s/ tasks in an ethical way	Students will be able to solve the various problems by using files, individually .	Students could be able to expose their knowledge of solving the problems using files by giving presentation and through discussion forum	Students could be able to continuously improve their knowledge in C concepts by implementing in emerging technologies.	The students will use the Knowledge on file for Data Science and Algorithms.	The students will use the knowledge of file handling for designing IoT & Mobile Application Developm ent.	The students will apply the file handling methods for developin g software projects in open-source environm ent.

- **Images / Screenshot of the practice:**

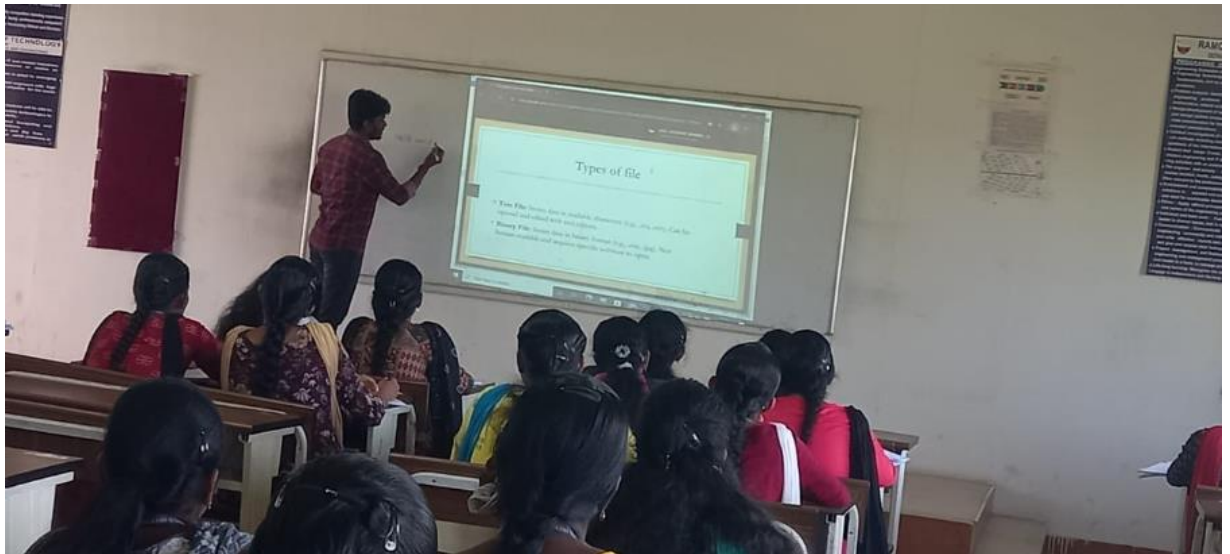


Figure 1. a Sequential Access Files explaining by M.Surulivelan



Figure 1. b Random Access Files explaining by M.Vinoth

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Some of the students understood the content that were delivered by the respective students.
- Some of them felt some style mismatches when compared to the previous sessions that were learned.

❖ ***Benefit of the practice:***

- Students access learning materials (e.g., videos, readings) before attending class.
- This approach provided more dynamic and interactive learning environment inside the classroom.
- It encourages the students to take ownership of their learning and actively participate in the learning process.
- This practice improves the presenting skills of the students.

❖ ***Challenges faced in implementation:***

- It made the presenters to spend more time outside the class room hours.
- Few of them couldn't understand the points delivered.
- After the presentation , the topics were discussed again and then all other students gained some good knowledge and idea about the topics.

References:

- <https://www.iberdrola.com/talent/flipped-classroom>
- <https://strobeeducation.com/blog/the-flipped-classroom-common-challenges/>
- <https://soeonline.american.edu/blog/flipped-classroom-pros-and-cons/>
- <https://www.ejmste.com/article/flipped-classroom-strategies-and-innovative-teaching-approaches-in-physics-education-a-systematic-13258>

Signature of Faculty Member

HOD